PACLAND

What it is, and why it matters — for people who have heard none of the story.

So what is it?

PACLAND is a verification discipline, applied one field at a time — physics first, then climate, economics, medicine. It does one thing relentlessly: it measures how much of a claim can actually be checked, and refuses the rest.

That used to be the whole story. It is not anymore. On top of verification we have now built a belief layer — the part that answers what verification cannot: how much should I believe each unproven claim, which one do I test next, and how honest has my confidence been?

The result is a clean, honest split for any claim, anywhere:

- exact — proven by a machine-checked proof, or counted exactly with no sampler and no run-to-run variance
- calibrated belief — a number with a stated margin, scored against how often it turned out right
- unknown — reported as unknown, never dressed up as a result

But how?

Three moves, and they transfer to any field:

- Ask for the constraints, the count and the deficit — never the values. A theory of everything predicts relations, never values.
- Make the computable part exact. 43,046,720 string-theory vacua counted exactly — twice, by two programs that share no code — no sampler, no probability, no run-to-run variance.
- Make the uncomputable part honest. Three gates stop a bad number reaching a decision; the belief layer scores what is still open.

The five results that changed the field

1 · “A theory of everything predicts relations, never values” — now a theorem

The exact content of the physics is constraints and labels, and nothing else. The tadpole is the unique invariant (machine-checked); the whole invariant ring is one generator. Masses, couplings and the cosmological constant are provably not exact.

2 · The Standard Model's charges are forced, not chosen

Anomaly cancellation pins the hypercharges uniquely (5/5), and three generations fall out as a topological index. The charge assignment we observe is the consistent one.

3 · The 19 floor — no completion derives the values

The 19 free parameters of particle physics are exactly the orbit-space dimension of the theory's own symmetry, and no tested completion derives more than two of them correctly.

4 · Deterministic in use

A field everyone assumed needed sampling returns an exact count — 43 million vacua, verified twice. The right question turns “intractable” into “exact”.

5 · The belief layer

What is still open is now handled honestly: calibrated belief, ranked by what to test next, with a probably-approximately-correct guarantee.

So what does it mean for research?

- Physics gets a falsifiable target: derive a value from a deeper theory, and the “relations, never values” thesis dies. Progress, with a way to be wrong.
- The method transfers: the same discipline already runs on climate, economics, medicine and — through Ask Alice — historical research. The physics results are the proof of concept; the discipline is the product.
- A new standard for uncertainty: “here is what is proven, here is what I believe and how calibrated that is” — a label nobody issues today, and the SDK behind it is live.

The bottom line

We did not solve string theory. We changed what it means to work on it: the exact part is now provable, and the open part is now a measured, calibrated, guaranteed belief instead of an adjective. That is the difference between believing and knowing.